

When Constants Win: Lessons from Re-analysing a Privacy-Harm Heuristics Evaluation

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Abstract

We re-analyse a small study that asked whether interpretable heuristics can estimate privacy harm, and how they compare with large language models (LLMs). Its recorded result was that a “rules static” arm beat a plain LLM, a rules-dynamic arm, retrieval augmentation, and two hybrids on all four reported metrics. Using only committed artifacts and no model calls, we find:

- Two constants chosen in advance, predicting *no harm* or *all four* of Solove’s harm groups for every case, match or beat every arm on exact match, Jaccard, and nDCG@5.
 - On micro-F1, the recorded arm leads, but no per-case outputs survive. A 2026 rerun of it is indistinguishable from the best constant unless out-of-taxonomy labels are removed.
 - In the committed code, five of the six arms sent an identical prompt.
 - The gold set was LLM-generated and mostly off-topic.
 - Every stored multi-arm experiment was a dry run on a different gold set.
 - Every training label came from a keyword fallback.
- We distil eight evaluation lessons and one data-ethics lesson.

1 Introduction

Estimating the *harm* of a privacy incident, not only detecting that one occurred, matters to regulators, auditors, and engineers deciding what to fix first. Solove’s taxonomy organises privacy problems into information collection, processing, dissemination, and invasion [22], and Citron and Solove catalogue the injuries that follow [4]. Recent work applies the taxonomy to AI incidents [11]. A natural question is whether cheap, interpretable heuristics can apply that structure at scale, and whether an LLM does better.

A 2025 master’s practicum asked exactly that: *can we estimate privacy harm using heuristics, given facts parsed by a small language model, and how do non-language models compare with LLMs and with an LLM applying classical privacy frameworks?* Its public artifact records a headline table in which a “rules static” arm led every metric. We re-examined every committed artifact offline and asked what that evidence supports. The contributions are:

- a re-analysis against pre-specified constant predictors, scored with the study’s own metric code, with paired bootstrap intervals for every gap (§3);
- an audit of what the evaluation actually compared: its

prompts, scorer revisions, gold set, dry runs, and label sources (§4);

- nine lessons, each tied to its evidence (§5).

2 The original study

Pipeline. Public incident reports (regulator and securities filings, news, forums, Wikipedia) were crawled, labelled with Solove categories, and featurised. Five interpretable model families were trained: a decision tree; an Explainable Boosting Machine [14, 15, 16]; Bayesian rule lists [12]; a sparse linear model; and a Chow–Liu Bayes net [3]. The labeller was to use GLiNER2 [24], a successor to GLiNER [23].

Evaluation. Six arms were scored on a 50-case gold set (the v3 set): a plain LLM baseline; *rules static*, a fixed rule block in the prompt; *rules dynamic*; RAG; and two hybrids. All were served by Gemini. The metrics were instance-averaged Jaccard, exact-match ratio (EMR), micro-F1, and nDCG@5 [25, 9] over Solove’s four groups. The recorded table is dated 2025-11-06 and reports means. The summary says there were five trials, but all six EMRs are multiples of 1/150, and three are not multiples of 1/250. That is consistent with three trials of 50 cases. A 2026 rerun repeated rules-static once. Its results file records `model_name: null` and no resolved model, so the model it used is known only from README prose. The repository also added an *offline* arm that calls no LLM and uses the corpus keyword labeller; we recompute it per case.

3 Constants against the arms

An empty-majority gold set. Of the 50 gold cases, 35 carry no harm label. The other 15 carry 35 group labels (processing 13, invasion 11, collection 7, dissemination 4). The scorer counts an empty prediction on an empty case as an exact match (EMR 1) but as a Jaccard and nDCG of 0. EMR therefore mostly measures how often an arm predicts nothing, and Jaccard and nDCG are read against a ceiling of 0.30 (Table 1).

Pre-specified constants. We fixed two label-free constants before looking at scores: the empty set, and all four groups in the taxonomy’s own order.

- *Always empty* scores EMR 0.700. That is higher than every arm, although the 2025 rules-static EMR (0.667) is within

Arm	All 50 cases					15 labelled cases	
	Jaccard	Jaccard ₀₌₁	EMR	micro-F1	nDCG@5	Jaccard	nDCG@5
<i>Recorded 2025 means (no per-case outputs survive)</i>							
Rules static	0.068	–	0.667	0.391	0.090	–	–
LLM baseline	0.056	–	0.620	0.285	0.072	–	–
Rules dynamic	0.055	–	0.593	0.281	0.073	–	–
Hybrid, deterministic first	0.053	–	0.593	0.275	0.068	–	–
RAG	0.047	–	0.580	0.242	0.061	–	–
Hybrid, LLM first	0.036	–	0.600	0.195	0.049	–	–
<i>Later runs (per-case outputs available)</i>							
Rules static, 2026, current scorer	0.105	0.465	0.360	0.336	0.176	0.351	0.587
same outputs, 2025 scorer	0.041	0.401	0.360	0.167	0.041	0.136	0.136
same outputs, filtered to groups (post hoc)	0.158	0.518	0.400	<i>0.463</i>	0.202	0.528	0.675
Offline keyword arm (no LLM)	0.081	0.161	0.080	0.222	0.162	0.269	0.541
<i>Constants, pre-specified</i>							
Always $\emptyset$	0.000	0.700	0.700	0.000	0.000	0.000	0.000
Always all four, taxonomy order	0.175	0.175	0.040	0.298	0.237	0.583	0.790
<i>Constants, best per column (selected on test labels)</i>							
Best of 65 constants	0.192	0.700	0.700	0.356	0.282	–	–
Perfect predictor (ceiling)	0.300	1.000	1.000	1.000	0.300	1.000	1.000

Table 1: All rows use the same 50-case gold set and the repository’s metric code, which reproduces the 2026 run exactly. That scorer gives empty-vs-empty a Jaccard and nDCG of 0 but an EMR of 1, so a perfect predictor scores only 0.300 on Jaccard and nDCG. Jaccard₀₌₁ uses the other common convention. The 65 constants are every ordered subset of the four groups. Bold marks the best value per column among the recorded arms, the 2026 current-scorer run, the offline arm, and the pre-specified constants. It excludes the post-hoc filtered row (italic where it would win), the test-selected constants, and the ceiling.

about two cases and cannot be tested without per-case outputs.

- *All four* beats every arm on Jaccard (0.175) and nDCG@5 (0.237), and so does each of its 24 orderings (nDCG 0.208–0.282).
- Paired case-resampling bootstrap intervals (2,000 resamples) against the 2026 rerun exclude zero: empty minus rerun EMR [0.22, 0.46]; all-four minus rerun Jaccard [0.033, 0.110] and nDCG [0.018, 0.111].
- Against the offline arm, all three of all-four’s gaps exclude zero: Jaccard [0.042, 0.152], micro-F1 [0.010, 0.149], nDCG [0.010, 0.146].
- Under the other Jaccard convention (empty-vs-empty = 1), the empty constant wins Jaccard as well. A constant wins either way; the convention decides which one.

Micro-F1, the exception. The recorded rules-static micro-F1 (0.391) exceeds even the best constant chosen with the test labels (0.356, {processing, invasion}), but no per-case outputs survive to test it. The 2026 rerun scores 0.336. Against the best constant re-selected inside each resample, the paired interval is [−0.086, 0.108]: no detectable difference. The rerun’s output contains eight labels outside the gold label space (root-cause terms such as `lack_of_consent`), and each is a guaranteed false positive. A post-hoc sensitivity check drops them. The filtered output reaches micro-F1 0.463, and its interval against the re-selected best constant is [0.005, 0.255]. So the arm carries signal that its prompt’s label vocabulary buries. Because the

filter was chosen after we saw the failure, this is a hypothesis for a new run, not a result.

The labelled cases alone. On the 15 non-empty cases, the all-four constant still leads on Jaccard (0.583 vs. 0.351 raw, 0.528 filtered) and nDCG@5 (0.790 vs. 0.587 and 0.675). On micro-F1 it roughly ties the filtered output (0.737 vs. 0.746). With only 15 informative cases, the micro-F1 intervals above are 0.19–0.25 wide, so modest real effects would go undetected [1].

Severity. The rerun also scored harm severity (1–5). Its accuracy, 0.50, is below that of always predicting the modal score 1 (0.78). Its quadratic-weighted κ , however, is 0.33 ([0.06, 0.57]) against 0 for the constant: this is where the arm shows clear signal. The gold severities are themselves inconsistent: four harm-labelled cases, such as “Data warehouse”, have severity 1. One unparseable prediction was scored as 0, outside the scale.

4 What the evaluation compared

Five arms, one prompt. In the committed code, the prompt differs by arm only for rules-static. The deterministic-first hybrid adds text only when a deterministic model’s output exists, and the runner never supplies one. The RAG arm receives no retrieved context. Building each arm’s prompt offline yields two distinct texts: one for rules-static, and one shared by the other five. Their spread (micro-F1 0.195–0.285, EMR 0.580–0.620) is therefore an estimate of replicate noise. Rules-static’s

micro-F1 lead over the best of them (0.106) is about the size of that spread (0.090). The recorded table predates the earliest committed runner (2025-11-24, in a private predecessor repository), which is built the same way, so this finding is strong but not conclusive. The “rules” are five generic compliance sentences, such as “If a data breach occurs, affected users must be notified within 72 hours.” They are unrelated to the trained models. Yet the original summary describes the arm as “deterministic regex/keyword heuristics”, although an LLM produced every one of its labels.

Scorer, prompt, and model changed together. The earliest committed runner mapped gold labels to the four groups but scored raw predicted labels. The current runner maps both. Scored the earlier way, the 2026 outputs fall to Jaccard 0.041, micro-F1 0.167, and nDCG 0.041: below the 2025 row on every metric. The prompt changed too. A revision on 2025-11-29 added a root-cause allowlist and told the model it may put root causes in the same field. The model is unrecorded in both runs, and the provider default in 2025 was already the model named in the 2026 README. The 2025→2026 differences are therefore a confound of scorer, prompt, possibly model, and one trial against a three-trial mean. They are not evidence of model drift. Hosted-model drift is real [2, 20], but this artifact cannot measure it.

An LLM-generated, mostly off-topic gold set. The summary records that the v3 gold set was generated with Gemini 2.5 Pro. Arms from the same family may share its errors. That is analogous to the self-preference of LLM judges [18, 26], and it would favour the arms, which makes the constants’ results more telling. Of the 50 cases, 47 are general Wikipedia articles; by title, none reports a specific privacy incident. The labelled cases mix on-topic articles (National Security Agency, Browser security, Profiling) with off-topic ones (the Railway Regulation Act 1844, Data warehouse). The JSON lists every title.

No case has the harms field the scorer reads first, so it silently falls back to `root_causes`. A code comment says that field holds technical causes, but in this set it holds only Solove subtypes, so the fallback read the right labels under the wrong name. The real contract failure is on the prediction side. The 2026 prompt invites root causes into the field the scorer treats as harms, which produced the eight out-of-taxonomy labels.

A second, 96-case set has labels drafted by `gemini-2.5-flash` (92 of 96) and is marked as reviewed in July 2026. Its review timestamps are bulk writes (median gap 1.4ms), so they cannot show per-case review. It has no empty-gold cases, which makes it the natural next benchmark, but no live comparison on it survives.

Dry runs stored as experiments. Fourteen “comprehensive” run directories, the largest spanning 320 configurations, all name `golden_cases.jsonl` (the file that later became the 96-case set), not the v3 set, so none bears on Table 1. Seven are flagged `dry_run`. The five that report per-arm metrics are all

dry runs, and within four of them every arm has an identical mean micro-F1; the fifth has two values. The placeholder score, 0.417, is higher than the headline arm’s 0.391, on a different gold set. The seven live runs committed no per-arm metrics. Pairwise-test files show Bonferroni thresholds and Cliff’s δ in three directories. One file is truncated and does not parse, and 15 of the 43 parseable entries are placeholders ($p = 1$, zero difference). A related failure path: on most API errors the LLM client returns a truncated echo of the prompt. That parses to an empty prediction, which earns EMR credit on 70% of cases. This is an unexamined alternative explanation for part of the high 2025 EMRs; the rerun contains one such failure.

Heuristic labels in, heuristic rules out. The labeller enables GLiNER2 only when no LLM provider is configured, and it silently swallows a GLiNER2 load failure, falling through to keyword scoring. In the released 33,472-row corpus, every label’s recorded source is that keyword fallback. 61.4% of labels are unknown, and 6 of the 21 other classes (e.g. `regulatory_violation`) are not Solove categories. The models were trained on seven features: a log count of affected individuals and six keyword flags. So they approximated one keyword heuristic from another, which cannot answer the study’s first question, and they did so poorly:

- Macro-F1 was 0.14 (decision tree), 0.10 (EBM), 0.20 (sparse linear), and 0.04 (Bayes net).
- The Bayes net predicted one class for every test row; that class makes up 52.4% of its test split but only 2.1% of the release. The release is therefore not the training distribution, and the training corpus’s recorded hash matches no archived copy we could find.
- The accuracies reduce to 43/82, 191/328, 289/492, and 28/45. A test split shared by all four models would need at least 14,760 rows, more than a 20% split of any corpus the repository documents, so the splits almost certainly differed.
- One of 16 per-category rule-list classifiers trained.
- The decision tree’s 20 exported “heuristics” all report support 0.000, yet its first rule (“no keyword present”) covers 61.4% of release rows. The training corpus is unavailable, but zero support for that rule is implausible and most likely an exporter bug.

5 Lessons

- L1 Pre-specify constant baselines and score them with the same scorer.** Use empty, all-labels in a fixed order, and the modal score for ordinal targets. Report oracle constants separately, re-selected inside the bootstrap. A metric on which no arm beats the pre-specified constants supports no claim.
- L2 Disclose empty-set conventions and the ceiling they imply.** Report the gold empty share, and score the non-empty subset separately.
- L3 Field names lie; pin the prompt–scorer contract.** Validate predictions and gold labels against one schema, and fail on missing fields or out-of-schema labels instead of

falling back.

- L4 Diff what each arm actually sends.** Arms that differ only in name measure replicate noise. That is useful, but only if you know it.
- L5 Make placeholders unable to pass for results.** Refuse to aggregate or test arm-invariant outputs, keep dry runs out of results directories, and fail on client errors instead of echoing the prompt.
- L6 Keep the gold set independent and its review auditable.** Do not generate it with the model family under test, and record who reviewed each case and when.
- L7 Measure the label pipeline you ran, not the one you designed.** Log the share of labels from each source, and never swallow a model-load failure [21, 17].
- L8 Record everything a comparison depends on, per result file.** That means the model version, scorer revision, prompt hash, trial count, and per-case outputs.

A data-ethics lesson. A pre-release audit found that the corpus’s social-media portions republished verbatim, re-identifiable posts in which private individuals described legal, medical, and financial trouble. That is the secondary-use harm the project studied, and a tested scrub removed it before release. Scrub at collection, not at release.

These lessons echo known failures: hypothesis-only and trivial baselines [8, 19], shortcut learning [7], leakage [10], fragile benchmark rankings [5], evaluation failures across ML [13], and significance testing [6]. What is new with LLMs is how cheaply one model family can produce each ingredient (gold labels, rules, predictions, and even the fallback text on error), and so end up on both sides of a comparison.

6 Limitations

- We made no model calls, so we cannot say how the arms would score on the 96-case set.
- The exact scorer and prompt behind the 2025 table are unknown.
- The constant-versus-2025 comparison is nonetheless robust:
 - the gold file is byte-identical to its Nov-2025 version;
 - the taxonomy that maps the gold set’s subtypes to groups is unchanged since the earliest committed runner, and the entries added later match no gold label;
 - constants emit group labels, so their scores do not depend on how predictions are mapped.
- If anything, the 2025 arms were handicapped by unmapped predictions. Mapping alone raises the rerun’s Jaccard from 0.041 to 0.105.
- The filtered sensitivity check is post hoc.
- Our judgement that the gold articles are off-topic rests on titles.
- Evidence from the private predecessor repository cannot be checked by readers.

7 Conclusion

Pre-specified constants match or beat every recorded arm on three of the four reported metrics. On the fourth, the recorded lead has no per-case evidence, and the rerun shows signal only after its out-of-taxonomy labels are removed. The study’s first question, whether heuristics can estimate harm, remains open, because every training label was itself a keyword heuristic. The uncertainty machinery (confidence intervals, corrected tests) ran only on dry runs against a different gold set, and the headline table carried no uncertainty at all. What the study offers is a compact, reproducible case of evaluation failure, and nine concrete ways to avoid repeating it.

Artifact availability

The repository, including gold sets, recorded runs, trained models, and a scrubbed corpus (release v1.0.0), is public at <https://github.com/Savage-Fred/privacy-harm-heuristics.paper/reanalysis.py> regenerates `paper/reanalysis.json`, the source of every re-analysis number, offline. It asserts that it reproduces the recorded 2026 rerun (set, ranking, and ordinal metrics) and the repository’s offline arm exactly.

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